

Incremental Economic Model Predictive Control for Grid-Connected PV-BESS Microgrids under Forecast and Model Uncertainty

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Abstract—The increasing penetration of photovoltaic generation and battery energy storage systems in grid-connected microgrids requires energy management strategies that can exploit forecast information while remaining effective under uncertainty. This paper investigates an incremental Economic Model Predictive Control strategy for a grid-connected PV-battery microgrid. The controller uses a **matrix-based predictor** and optimizes battery-power increments over a receding horizon, **while a nonlinear simulated plant is deliberately separated from the internal control model.** This structure enables simultaneous assessment of irradiance and load forecast errors together with mismatch in PV rating, battery capacity, efficiencies, and power limits. A reproducible seven-day closed-loop study compares the proposed strategy with a no-storage baseline and a perfect-information, matched-model benchmark. Despite the combined uncertainty, grid imports are reduced by 83.1% and peak import by 10.6% relative to the no-storage case, while the plant state of charge remains within the target operating interval. The economic result remains within approximately 9.5% of the ideal predictive benchmark. These results highlight the value of receding-horizon feedback under model and forecast mismatch and provide a reproducible baseline for subsequent hard-constrained formulations and validation with measured campus data.

Index Terms—battery energy storage system, economic model predictive control, energy management system, microgrid, photovoltaic generation, predictive control

I. INTRODUCTION

The growing penetration of distributed renewable energy resources is accelerating the deployment of microgrids that combine local generation, energy storage, controllable demand,

and grid interaction. Their operation, however, is complicated by the variability of photovoltaic (PV) generation and load demand, together with electricity-price signals and technical constraints. Energy management systems (EMSs) are therefore required to coordinate these resources in an efficient and reliable manner [1], [2].

Battery energy storage systems (BESSs) add temporal flexibility by decoupling generation and consumption. In grid-connected applications, storage can increase local renewable-energy utilization, reduce peak grid demand, and shift energy exchange toward more favorable periods [3]. These benefits depend strongly on the charge/discharge schedule because each battery action changes the energy available at future instants. Consequently, purely reactive control rules cannot explicitly exploit forecasts of PV generation, load demand, and electricity prices.

Model Predictive Control (MPC) is well suited to this class of problems because it combines a predictive model, an optimization objective, operating constraints, and repeated state feedback over a receding horizon. At the grid-management level, forecasts of load, renewable generation, and electricity prices can be incorporated directly into the decision process [2], [4]. Economic MPC (EMPC) extends this concept by optimizing economic or operational criteria rather than tracking a fixed reference. Recent surveys show that EMPC has been applied to stand-alone, grid-connected, residential, and networked microgrids using deterministic, stochastic, centralized, distributed, and hybrid formulations [5], [6].

A second challenge is the transition from numerical scheduling to controllers that remain effective under realistic uncertainty and implementation constraints. Field-oriented linear

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EMPC has been demonstrated using industrial information and communication infrastructure [7], while hierarchical stochastic MPC has been experimentally validated with forecast errors explicitly incorporated into the control design [8]. Recent work on a PV-battery testbench combined neuro-fuzzy forecasting with real-time EMPC implemented through a PLC and a Python supervisory layer, showing close agreement between simulation and experiment and improved economic performance relative to a hysteresis-based strategy [9]. The same experimental platform has also been extended to predictive management of stationary storage and bidirectional electric vehicles [10].

These developments also expose an important validation issue. The performance of predictive energy management depends not only on the optimizer, but also on forecast quality and on the agreement between the internal controller model and the physical plant. Receding-horizon feedback mitigates disturbances by repeatedly updating the optimization with new measurements [2]; nevertheless, experimental implementations still exhibit deviations caused by unmodeled losses, efficiency variations, and other plant-model discrepancies [9]. Evaluating an EMS only with perfect forecasts and a plant identical to its prediction model can therefore overestimate the performance expected in closed-loop operation.

MPC-based energy management has also been investigated in the Brazilian context. A hybrid MPC formulation incorporating Brazilian energy-market rules was proposed for renewable microgrids in [11], while experimental work at UFSC demonstrated economic management of a laboratory microgrid integrating renewable generation, hybrid storage, grid interaction, and vehicle-to-grid resources [12]. These results provide a relevant foundation for scalable predictive-control approaches aimed at larger PV-BESS applications.

Against this background, this paper evaluates an incremental EMPC for a grid-connected PV-BESS microgrid using an explicit separation between the controller prediction model and a nonlinear simulated plant. Forecast errors and deliberate parameter mismatch are introduced simultaneously, and the resulting closed-loop behavior is compared with both a no-storage baseline and a perfect-information, matched-model predictive benchmark. The implemented controller uses **matrix-based prediction** and Sequential Least Squares Programming (SLSQP), establishing a reproducible baseline before subsequent hard-constrained reformulations.

The main contributions are summarized as follows:

- an incremental economic MPC formulation in which battery-power increments are optimized over a finite control horizon;
- an explicit controller-plant separation that enables simultaneous analysis of forecast uncertainty and parameter mismatch;
- a reproducible closed-loop comparison with no-storage and perfect-information benchmarks using identical real-ized profiles; and
- a quantitative assessment based on grid exchange, peak demand, battery utilization, SOC evolution, and economic

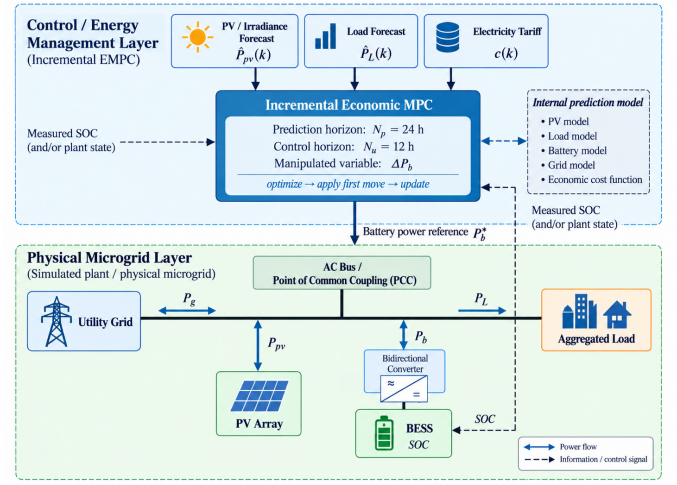


Fig. 1. Two-layer architecture of the grid-connected PV-BESS microgrid and the incremental economic MPC. The upper layer corresponds to the control/energy-management level, which receives forecasts and measured plant information and computes the battery-power reference in a receding-horizon fashion. The lower layer represents the physical microgrid, including the utility grid, AC bus/PCC, PV array, bidirectional converter, BESS, and aggregated load.

performance.

The remainder of this paper is organized as follows. Section II describes the PV-BESS model. Section III presents the incremental EMPC formulation. Section IV describes the validation methodology and uncertainty scenarios. Section V discusses the results, and Section VI summarizes the main conclusions.

II. PV-BESS MICROGRID MODEL

A. System Architecture

The considered system is a grid-connected AC microgrid composed of an aggregated electrical load, a PV generation system, a BESS, and the external utility grid. At each sampling instant k , the power balance is

$$P_g(k) + P_{pv}(k) + P_b(k) = P_L(k), \quad (1)$$

where P_g is the grid power, P_{pv} is the available photovoltaic power, P_L is the electrical load, and P_b is the battery power.

The sign convention adopted in this work is $P_b > 0$ for battery discharge and $P_b < 0$ for battery charge. Likewise, $P_g > 0$ represents grid import, while $P_g < 0$ denotes export. From (1),

$$P_g(k) = P_L(k) - P_{pv}(k) - P_b(k). \quad (2)$$

Equation (2) directly couples the BESS control action to the power exchanged with the utility grid.

B. Battery Energy Model

For energy-management purposes, the battery is represented by a discrete-time stored-energy model. Defining $E_b(k)$ as the battery energy at instant k , the nominal prediction model is

$$E_b(k+1) = E_b(k) - T_s P_b(k), \quad (3)$$

where T_s is the sampling interval.

The corresponding SOC is

$$SOC(k) = \frac{E_b(k)}{E_{\text{nom}}}, \quad (4)$$

where E_{nom} is the usable battery-energy capacity assumed by the controller.

For closed-loop plant simulation, charging and discharging efficiencies are explicitly separated:

$$E_b^p(k+1) = E_b^p(k) + T_s \left[\eta_c P_{\text{ch}}(k) - \frac{P_{\text{dis}}(k)}{\eta_d} \right], \quad (5)$$

$$P_b(k) = P_{\text{dis}}(k) - P_{\text{ch}}(k), \quad (6)$$

where $P_{\text{ch}} \geq 0$, $P_{\text{dis}} \geq 0$, and η_c and η_d denote the effective charging and discharging efficiencies.

This distinction allows the prediction model and the plant model to be intentionally different during robustness tests.

The SOC operating region is bounded by

$$SOC_{\min} \leq SOC(k) \leq SOC_{\max}, \quad (7)$$

with nominal values

$$SOC_{\min} = 0.10, \quad SOC_{\max} = 0.90. \quad (8)$$

Battery power is constrained by

$$-P_{b,\max}^{\text{ch}} \leq P_b(k) \leq P_{b,\max}^{\text{dis}}, \quad (9)$$

The controller model used in the present simulation assumes

$$E_{\text{nom}}^{\text{MPC}} = 900 \text{ kWh}, \quad -300 \leq P_b^{\text{MPC}} \leq 300 \text{ kW}. \quad (10)$$

The simulated plant intentionally uses different parameters to assess model mismatch. In the repository baseline, the plant battery capacity is 874 kWh, its charge/discharge power limits are -294/314 kW, and its charge/discharge efficiencies are 0.50/1.00, whereas the controller assumes 0.95/0.95. These values are treated as a synthetic stress-test configuration rather than as parameters of a specific commercial BESS.

C. Incremental Control Input

Rather than directly optimizing an unconstrained battery-power sequence, the proposed formulation uses the battery-power increment

$$\Delta P_b(k) = P_b(k) - P_b(k-1) \quad (11)$$

as the manipulated variable. Therefore,

$$P_b(k) = P_b(k-1) + \Delta P_b(k). \quad (12)$$

Over the prediction horizon, the future battery-power vector can be expressed as

$$\mathbf{P}_b = \mathbf{1}P_b(k-1) + \mathbf{T}\Delta\mathbf{P}_b, \quad (13)$$

where $\mathbf{T}$ is a lower-triangular accumulation matrix and

$$\Delta\mathbf{P}_b = [\Delta P_b(k) \quad \cdots \quad \Delta P_b(k+N_p-1)]^T. \quad (14)$$

The incremental formulation introduces an explicit mechanism for limiting command variations,

$$-\Delta P_{b,\max} \leq \Delta P_b(k+i) \leq \Delta P_{b,\max}, \quad (15)$$

thereby reducing abrupt transitions between consecutive BESS setpoints.

D. Grid Constraints

The microgrid connection can also be subject to power limits,

$$-P_{g,\max}^{\text{exp}} \leq P_g(k) \leq P_{g,\max}^{\text{in}}, \quad (16)$$

Using (2), these limits can be directly mapped into constraints on the predicted battery trajectory. If energy export is not remunerated, the controller can additionally impose

$$P_g(k) \geq 0, \quad (17)$$

or alternatively use independent import and export tariffs when bidirectional exchange is allowed.

III. INCREMENTAL ECONOMIC MODEL PREDICTIVE CONTROL

A. Receding-Horizon Formulation

At each control instant k , forecasts are available for the following N_p intervals. The predicted net load is defined as

$$\hat{P}_n(k+i|k) = \hat{P}_L(k+i|k) - \hat{P}_{\text{pv}}(k+i|k). \quad (18)$$

The corresponding predicted grid power is

$$\hat{P}_g(k+i|k) = \hat{P}_n(k+i|k) - P_b(k+i|k). \quad (19)$$

A prediction horizon of

$$N_p = 24 \quad (20)$$

is adopted with hourly sampling, $T_s = 1$ h, thereby representing a 24-h look-ahead window. The control horizon is selected as

$$N_u = 12. \quad (21)$$

After optimization, only the first BESS control action is applied to the plant. At the following sampling instant, the measured plant SOC and updated forecasts are used to solve a new optimization problem.

B. Economic Objective Function

The primary objective is the minimization of the predicted grid-energy cost over the horizon. When import and export prices are represented independently, positive and negative grid powers are defined as

$$P_g = P_g^+ - P_g^-, \quad P_g^+ \geq 0, \quad P_g^- \geq 0. \quad (22)$$

The grid cost is then

$$J_g = \sum_{i=0}^{N_p-1} [c_{\text{buy}}(k+i)P_g^+(k+i) - c_{\text{sell}}(k+i)P_g^-(k+i)] T_s. \quad (23)$$

Battery operation is regularized through the quadratic penalty

$$J_b = \sum_{i=0}^{N_b-1} P_b^2(k+i|k), \quad (24)$$

while changes in battery power are penalized using

$$J_\Delta = \lambda_\Delta \sum_{i=0}^{N_u-1} \Delta P_b^2(k+i|k). \quad (25)$$

The terminal stored-energy objective is

$$J_f = \lambda_{\text{SOC}} [SOC(k+N_p|k) - SOC_f]^2. \quad (26)$$

The complete objective function is

$$J = J_g + J_b + J_\Delta + J_f. \quad (27)$$

The implemented controller solves the finite-horizon nonlinear constrained optimization using Sequential Least Squares Programming (SLSQP). The optimized decision vector is the sequence of battery-power increments

$$\Delta \mathbf{P}_b = [\Delta P_b(k) \quad \cdots \quad \Delta P_b(k+N_u-1)]^T. \quad (28)$$

For each candidate sequence, the matrix predictor reconstructs the virtual battery-power trajectory, predicted SOC, PV generation, and grid power. The objective combines the predicted economic grid cost with a quadratic penalty on the control increments and a soft penalty for predicted SOC excursions,

$$J = J_g + \lambda_\Delta \sum_{i=0}^{N_u-1} \Delta P_b^2(k+i|k) + \rho_{\text{SOC}} [SOC(k+N_p|k) - SOC_{\text{max}}]^2, \quad (29)$$

where J_{SOC} penalizes predicted values below SOC_{min} or above SOC_{max} . In the repository implementation, SLSQP is configured with a convergence tolerance of 10^{-9} and a maximum of 200 iterations.

C. Operational Constraints and Soft SOC Penalty

The implemented controller directly constrains the battery-power increments and the resulting virtual battery power. The increment is bounded by

$$-\Delta P_{b,\text{max}} \leq \Delta P_b(k+i) \leq \Delta P_{b,\text{max}}, \quad (30)$$

with $\Delta P_{b,\text{max}} = 100$ kW in the current test case. The cumulative power trajectory is constrained within the controller limits of ± 300 kW.

In contrast, the current repository implementation does not impose predicted SOC as a hard linear constraint inside SLSQP. Instead, violations of the controller interval $0.1 \leq SOC \leq 0.9$ are penalized quadratically with a weight of 10^7 . The simulated plant independently enforces its own physical SOC interval by saturating the realizable charge/discharge power before updating the battery state.

This distinction is important for interpreting the results: battery-power and increment bounds are hard optimization constraints, whereas the controller-side SOC limits are soft constraints in the present implementation. A hard-constrained QP reformulation is therefore considered a direct extension of the baseline developed here.

D. Nonlinear MPC Benchmark

A nonlinear optimization-based MPC using Sequential Least Squares Programming (SLSQP) is retained as a benchmark because it represents the original numerical implementation from which the present formulation is developed.

Both controllers use equivalent forecasts, sampling intervals, main operating constraints, and economic objectives whenever possible. The essential difference lies in the numerical structure: the SLSQP implementation evaluates a general constrained nonlinear optimization problem, whereas the proposed controller exploits the linear battery model and quadratic objective to assemble a convex QP directly.

The comparison therefore investigates whether the convex formulation preserves the economic behavior of the original controller while providing more predictable constraint handling and numerical performance.

IV. CLOSED-LOOP VALIDATION METHODOLOGY

A. Simulation Horizon and Nominal Parameters

The controller is evaluated in closed loop over a representative multi-day operating horizon. A simulation interval of

$$N_s = 168 \text{ h} \quad (31)$$

is considered, corresponding to seven days with hourly sampling.

The nominal PV installation is rated at approximately

$$P_{\text{pv,r}} = 400 \text{ kW}. \quad (32)$$

The controller assumes a BESS energy capacity of 900 kWh and a power limit of ± 300 kW to evaluate model mismatch, the simulated physical plant can instead use

TABLE I
REPOSITORY BASELINE PARAMETERS

Parameter	Plant	MPC model
Sampling period T_s	1 h	1 h
Simulation duration	168 h	–
Prediction horizon N_p	–	24
Control horizon N_u	–	12
PV nominal power	400 kW	425 kW
BESS energy capacity	874 kWh	900 kWh
Charge efficiency	0.50	0.95
Discharge efficiency	1.00	0.95
Charge power limit	294 kW	300 kW
Discharge power limit	314 kW	300 kW
SOC interval	0–1	0.1–0.9 (soft)
Initial SOC	0.48	measured feedback
$\Delta P_{b,\max}$	–	100 kW
λ_Δ	–	5×10^{-9}
SOC penalty weight	–	10^7

$$E_{b,\text{plant}} = 874 \text{ kWh}, \quad (33)$$

while the controller continues operating with the nominal 900 kWh model. Additional mismatch cases may be obtained by modifying the effective charging and discharging efficiencies.

B. Synthetic Forecast and Disturbance Generation

The repository uses synthetic profiles to create a controlled difference between the trajectories supplied to the MPC and those experienced by the plant. For the reproducible results reported in this paper, the random generator is initialized with seed 42.

The irradiance forecast follows a daylight sinusoidal profile with a daily amplitude randomly drawn around 800 W/m². The realized irradiance is obtained by multiplying the forecast profile by an hourly Gaussian factor with mean 1 and standard deviation 0.2, while nighttime irradiance is forced to zero. For the selected realization, the daylight irradiance error presents a mean absolute error of 63.4 W/m² and an RMSE of 92.0 W/m².

The synthetic load is obtained from a repeated 24-h profile ranging from approximately 12.5 to 42 kW, with additive zero-mean Gaussian noise of 2 kW standard deviation. The resulting load forecast error has a mean absolute error of 1.59 kW and an RMSE of 2.02 kW.

The economic signal is also synthetic. The purchase tariff is fixed at R\$ 3.00/kWh, while the export tariff is R\$ 1.98/kWh from 18:00 to 21:00 and R\$ 0.58/kWh during the remaining hours. Over the seven-day realization, the plant produces 14.88 MWh of PV energy while the load consumes 4.33 MWh. The experiment is therefore intentionally export-dominant and should be interpreted as a stress-test scenario rather than as a representation of the present UFSC campus sizing or tariff.

Fig. 2 condenses forecast and plant uncertainty into the net-load signal directly relevant to grid exchange. The prediction uses the controller PV rating, whereas the realized trajectory uses the plant PV rating and disturbed irradiance/load profiles.

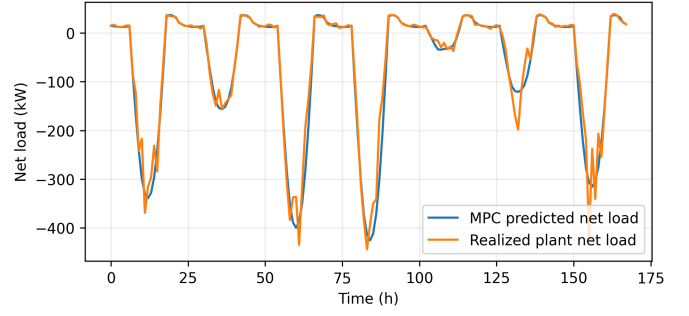


Fig. 2. Predicted and realized net-load trajectories for the reproducible seven-day uncertainty realization.

C. Controller–Plant Separation

The validation framework explicitly distinguishes the prediction model used internally by the EMPC from the plant model used to propagate the actual BESS state.

At time k , the measured plant state $SOC^p(k)$ is provided to the controller. The optimization generates the sequence

$$P_b^*(k|k), \dots, P_b^*(k + \overset{N_u}{\underset{N_p}{N_p}} - 1|k). \quad (34)$$

Only the first element, $P_b^*(k|k)$, is applied to the plant. The physical plant then evolves according to (5), using the actual capacity and efficiency parameters. At $k + 1$, the resulting measured state $SOC^p(k + 1)$ is fed back to the controller and the complete optimization is repeated.

This approach prevents accumulated prediction errors from remaining hidden inside a purely open-loop scheduling result.

D. Evaluated Cases

Three cases are evaluated using the same realized irradiance and load trajectories:

- **C0 – No BESS:** the battery is disabled, $P_b(k) = 0$, establishing the grid-exchange and economic baseline.
- **C1 – Perfect-information matched-model MPC:** the predictive controller receives the realized future irradiance and load profiles and uses the same PV and battery parameters as the simulated plant. This case provides an ideal predictive benchmark for the implemented controller structure.
- **C2 – Uncertain/mismatched incremental MPC:** the controller receives the synthetic forecast profiles and uses the mismatched internal parameters listed in Table I. This is the principal case corresponding to the uncertainty structure of the project repository.

All three cases use the same realized plant profiles generated with seed 42, allowing differences in performance to be attributed to storage operation and predictive information rather than to different stochastic realizations.

E. Performance Metrics

Total grid energy cost is computed from the actual plant-side grid exchange,

$$C_g = \sum_{k=1}^{N_s} [c_{\text{buy}}(k)P_g^+(k) - c_{\text{sell}}(k)P_g^-(k)] T_s. \quad (35)$$

The relative cost reduction with respect to the no-storage case is

$$\Delta C = 100 \frac{C_j - C_0}{C_0}. \quad (36)$$

Imported and exported energies are

$$E_{\text{imp}} = \sum_k P_g^+(k) T_s, \quad (37)$$

$$E_{\text{exp}} = \sum_k P_g^-(k) T_s. \quad (38)$$

Peak grid import is

$$P_{\text{peak}} = \max_k P_g(k), \quad (39)$$

and battery throughput is

$$E_{\text{thr}} = \sum_k |P_b(k)| T_s. \quad (40)$$

An equivalent full-cycle indicator is obtained from

$$N_{\text{eq}} = \frac{E_{\text{thr}}}{2E_{\text{nom}}}. \quad (41)$$

Constraint compliance is evaluated from the minimum and maximum actual SOC and from the number and magnitude of power or SOC violations. Solver performance is characterized by average solution time, 95th-percentile solution time, maximum solution time, number of successful optimizations, number of infeasible problems, and number of numerical solver failures.

V. RESULTS AND DISCUSSION

A. Closed-Loop Battery Operation

Fig. 3 shows the measured plant SOC obtained with C2. Although the controller treats the interval $0.1 \leq \text{SOC} \leq 0.9$ through a quadratic soft penalty rather than a hard optimization constraint, the realized SOC remains between 10.01% and 78.56% throughout the seven-day experiment. The final SOC is 26.98%. No physical SOC or battery-power saturation event is activated by the nonlinear plant model in this realization.

The BESS exchanges approximately 8.99 MWh of absolute energy throughput during the week, corresponding to about 5.14 equivalent full cycles when normalized by the 874-kWh plant capacity. The maximum charging power is 147.7 kW and the maximum discharging power is 98.8 kW, both below the physical plant limits.

Fig. 4 shows the first 72 h of the realized power trajectories. The large PV-to-load ratio produces substantial export during daytime periods, while the predictive controller shifts part of this energy through the BESS in response to the time-varying export tariff.

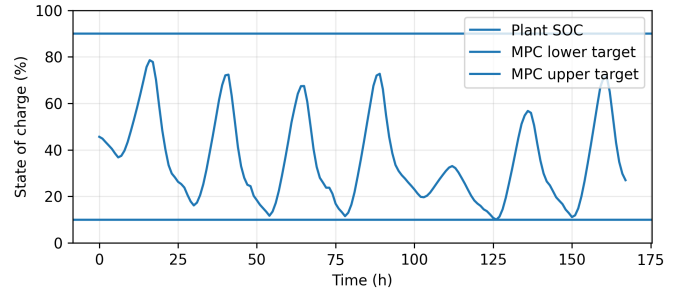


Fig. 3. Measured plant SOC for C2. Horizontal lines indicate the lower and upper SOC levels used by the MPC penalty.

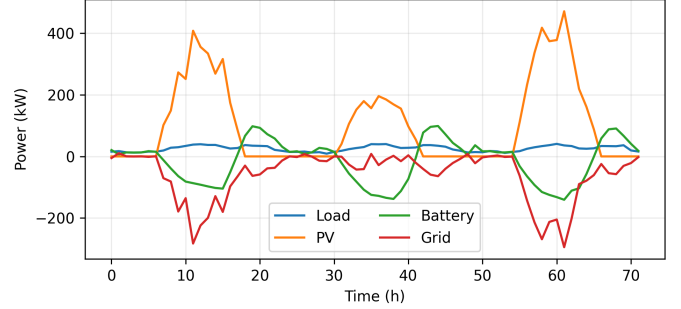


Fig. 4. Representative 72-h plant-side power trajectories for C2. Positive battery power denotes discharge and positive grid power denotes import.

B. Energy and Economic Performance

Table II compares the no-storage baseline, the perfect-information benchmark, and the uncertainty-and-mismatch case. With no BESS, the microgrid imports 1.96 MWh and exports 12.51 MWh. The uncertain MPC reduces imported energy to 0.331 MWh, corresponding to an 83.1% reduction, while peak import decreases by 10.6%, from 39.31 to 35.13 kW.

The perfect-information matched-model benchmark further reduces imported energy to 0.165 MWh and peak import to 23.89 kW. This gap quantifies the combined effect of forecast error and model mismatch under the selected synthetic scenario.

Because exported energy is remunerated in the synthetic tariff, the net operating cost becomes negative in all three cases; negative values therefore represent net revenue rather than an expense. For this reason, a conventional percentage cost reduction is not an appropriate metric. Instead, the import cost and export revenue are reported separately.

C2 changes the weekly net balance from R\$ -1375.18 without storage to R\$ -5107.34. The perfect-information benchmark reaches R\$ -5643.29. Expressed as net revenue, C2 retains approximately 90.5% of the ideal C1 result despite simultaneous forecast and model uncertainty.

Fig. 5 shows the cumulative economic balance. The principal contribution of the BESS is not an increase in total exported energy; storage losses reduce exported energy relative to C0. Instead, the controller temporally reallocates energy

TABLE II
SEVEN-DAY CLOSED-LOOP PERFORMANCE FOR THE REPRODUCIBLE SEED-42 SCENARIO

Case	E_{imp} (kWh)	E_{exp} (kWh)	Import cost (R\$)	Export rev. (R\$)	Net cost (R\$)	$P_{\text{imp}}^{\text{max}}$ (kW)	$P_{\text{exp}}^{\text{max}}$ (kW)	E_{thr} (kWh)
C0 – No BESS	1960.29	12510.43	5880.87	7256.05	-1375.18	39.31	444.26	0
C1 – Perfect/matched MPC	164.76	7789.82	494.29	6137.59	-5643.29	23.89	258.96	9532.05
C2 – Uncertain/mismatched MPC	331.05	8130.50	993.15	6100.49	-5107.34	35.13	317.66	8986.81

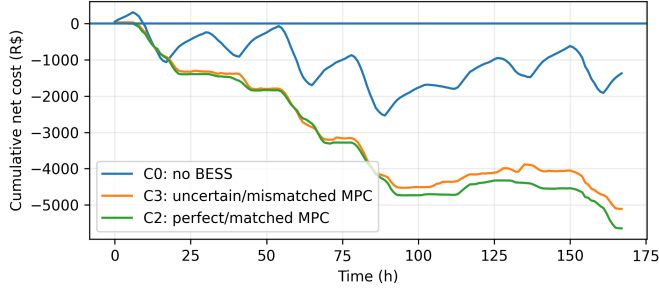


Fig. 5. Cumulative net operating cost for C0, C1, and C2. Negative values represent net revenue.

and substantially reduces expensive grid imports while shifting export toward more favorable periods.

C. Effect of Controller–Plant Mismatch

The C2 controller assumes a 900-kWh battery and a 425-kW PV array, whereas the simulated plant uses 874 kWh and 400 kW. More importantly, the controller assumes symmetric charge/discharge efficiencies of 0.95/0.95, while the stress-test plant uses 0.50/1.00. The latter discrepancy is intentionally severe and should not be interpreted as a representative commercial BESS efficiency.

Despite these differences, measured-state feedback prevents the SOC prediction error from accumulating indefinitely. At every sampling instant, the plant SOC is reintroduced as the initial condition of the next MPC problem. The resulting C2 trajectory remains inside the controller SOC penalty interval in the reproduced experiment, but its 9.5% net-revenue gap relative to C1 demonstrates that repeated feedback does not eliminate the economic effect of inaccurate forecasts and model parameters.

The current result therefore supports two distinct conclusions. First, explicit controller–plant separation is necessary when assessing predictive energy management under uncertainty. Second, the present soft-SOC SLSQP formulation provides a useful baseline, but it does not yet establish guaranteed hard constraint satisfaction. A hard-constrained QP formulation remains a relevant extension for the final controller architecture.

VI. CONCLUSION

This paper presented an incremental economic MPC framework for the energy management of a grid-connected PV-BESS microgrid with explicit separation between the controller prediction model and a nonlinear simulated plant. The

implemented controller optimizes battery-power increments over a 24-h prediction horizon and a 12-h control horizon using SLSQP, while battery-power and increment limits are directly constrained and predicted SOC excursions are handled through a quadratic soft penalty.

In the reproducible seven-day uncertainty scenario, the controller reduced imported grid energy from 1.96 MWh without storage to 0.331 MWh and reduced peak import from 39.31 to 35.13 kW. The measured plant SOC remained between 10.01% and 78.56% without activating physical plant saturation. Under the export-dominant synthetic tariff, the net operating balance improved from R\$ -1375.18 without storage to R\$ -5107.34, while a perfect-information matched-model benchmark reached R\$ -5643.29. Thus, the uncertain/mismatched controller retained approximately 90.5% of the ideal benchmark net revenue.

These results establish a reproducible numerical baseline and show the relevance of evaluating predictive BESS control with plant-model mismatch and forecast uncertainty rather than only under nominal open-loop conditions. Future work will introduce explicit hard SOC constraints, a convex QP implementation, rule-based benchmarking, multiple paired stochastic realizations, and validation with measured campus load, PV, and tariff data.

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